

Problem 49. A hotel has 100 rooms with capacities for 101, 102, $\dots$, 200 people, respectively. These rooms are occupied by n people in total. Banach arrives, and Hilbert, the owner, wants to provide him with a personal room. For that purpose, Hilbert chooses two rooms, A and B , and moves all the people from room A to room B without exceeding its capacity. Determine the largest n for which Hilbert can be sure he is able to reach the goal whatever the initial distribution of people is.

Problem 50. Suppose that a competition has 300 participants. For any 3 participants there are 2 that are not friends. Euler enumerates the participants in some order and denotes by x_i the number of friends of the i th participant. If

$$\{x_1, x_2, \dots, x_{299}, x_{300}\} = \{1, 2, \dots, N-1, N\},$$

find the largest possible value of N .

Problem 51. Consider an equilateral triangular array of numbers consisting of 2019 rows with k numbers in the k th row, $k = 1, 2, \dots, 2019$. Except for the numbers in the bottom row, each number is equal to the sum of two numbers immediately below it. How many ways can one assign 0 or 1 to each of the numbers $a_0, a_1, \dots, a_{2018}$ (from left to right) in the bottom row such that the number S on the top is divisible by 1009?

Problem 52. A sequence $\{a_n\}_{n \geq 1}$ of real numbers is defined as follows:

$$a_1 = 2, \quad a_{n+1} = \frac{a_n^2 + 1}{2}, \text{ for } n \geq 1.$$

Prove that

$$\sum_{j=1}^N \frac{1}{a_j + 1} < 1$$

for every $N = 1, 2, \dots$